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Christian Veddeler
Joran Kuijper
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(Eds.)

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Future Cities—City Futures
Emerging Urban Perspectives

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COLOPHON

Future Cities—City Futures
Emerging Urban Perspectives

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ABSTRACT

Although it may seem intuitive that people prefer spaces that are occupied to some extent and at other times are concerned by overcrowding, the tipping points at which a space becomes too empty or too crowded are difficult to define. In architecture and planning, new computational approaches such as space syntax and spatial network analysis allow designers to forecast aggregate levels of pedestrian flow and occupancy with remarkable accuracy. But is there really a single optimum occupancy? How does the design and aesthetic of spaces change our perceptions? These are some of the underlying questions motivating our research project, a Neuroscientific investigation of the Interaction between Crowdedness and Environment typology. Neuroscientific methods such as mobile electroencephalography have in recent years begun to change how we study the spontaneous and unconscious responses of people to their environment. In this chapter, we weave together a conceptual overview of how people perceive the environment, classic and contemporary approaches to crowding, and how neuroscience aspires to change urban planning.

KEYWORDS

urban design; social density; aesthetics; environmental neuroscience.

Chapter 8—Disentangling the Social from the Aesthetic: Neuropsychology for Urban Design

Panagiotis Mavros, Agnieszka Olszewska-Guizzo, and Dominique Makowski

Cities changed during the pandemic. Because of movement restrictions and lockdowns, downtown streets that used to bustle with crowds stood empty. Even today, business districts feel different because people have switched to working from home.

INTRODUCTION

In some cities, the street space was reclaimed from cars and given to cafés and restaurants to set up outdoor tables at which their patrons could congregate more safely. These streets have changed too; now they are occupied by furniture instead of parking spaces, and most importantly, people can be seen at various times of day.

It is said that the sloped public square in front of the Centre Pompidou in Paris resembles a beach; it looks as welcoming when it is empty of people as when it is full. Indeed, no matter what time of the day one walks by, the space can be appreciated whether or not people are sitting, playing music, or watching impromptu performances. However, other

places lose their appeal when they are too empty or too full of people; some streets feel overwhelming and exhausting, while others feel dull or unsafe. Can design features and aesthetics moderate the effect of occupancy, or of crowding, on how spaces are perceived?

Although it may seem intuitive that people prefer spaces that are occupied to some extent and at other times are concerned by overcrowding, the tipping points at which a space becomes too empty or too crowded are difficult to define. In architecture and planning, new computational approaches such as space syntax and spatial network analysis allow designers to forecast aggregate levels of pedestrian flow and occupancy with remarkable accuracy. But is



Figure 1 ► p. 151

there really a single optimum occupancy? How do the design and aesthetic of spaces change our perceptions?

These are some of the questions motivating our new joint research project, a Neuroscientific investigation of the Interaction between Crowdedness and Environment typology (NICE). Neuropsychology, the science of the relationship between mental activity and its brain substrates, is a relevant and promising approach to investigating the role of architecture and urban design in people's lives. The application of neuroscientific methods such as mobile electroencephalography has recently begun to change how we study the spontaneous and unconscious responses of people to their environments. In this chapter, we weave together a conceptual overview of how people perceive the environment, classic and contemporary approaches to crowding, and how neuroscience aspires to change urban planning.

REDEFINING URBAN VIBRANCY

From Jane Jacobs to William H. Whyte, Jan Gehl, and others, theorists and practitioners of urban design have also realized the multiplier effect of social presence for feelings of safety as well as for urban vibrancy and enjoyment. Interestingly, this social turn can be observed in recent trends in architectural visualization. However, we often use the term 'crowding', which carries negative connotations, across various domains of life from the depletion of attentional and cognitive resources to feelings of discomfort and stress, from housing (Evans et al., 2007) to public transport (Haywood et al., 2015), and even in retail and tourism (Yin et al., 2020). As a result, interest in the psychological effects of crowding on people is longstanding and often driven by residential and urban crowding. Early studies showed that higher social density reduces cognitive performance and elevates psychological and physiological stress (Saegert, 1973; Evans, 1979); more recent studies have shown that crowding can even skew perceptions of social class and product

valuation (Baker & Wakefield, 2012). In such approaches, crowding is intrinsically viewed as something negative that has to be avoided through scheduling or design and planning. This is analogous to traffic in transport studies, which find that increasing the throughput of highways leads to increased driver satisfaction. But people walking in urban space are not the same as car drivers idling at a traffic junction.

The pandemic perhaps attuned us to the fact that social presence is also necessary for urban vibrancy. Indeed, in line with Hebb's (1955) theory of optimal arousal, there seems to be an inverse 'U' relationship between levels of crowding and willingness to visit urban public spaces (Nouri, 2011). Indeed, both our intuition and empirical studies suggest that a degree of social density is desirable and indeed even beneficial. This is the case for several social animals too: studies have shown that too little and too much social interaction are both detrimental for health (Love & Zelikowsky, 2020). In other words, moderate levels of social density are preferable, and spaces that are too empty or too full become less attractive.

There might be a difference between objective and subjective experiences of crowding. In the 1970s, researchers began distinguishing between the spatial condition of density and the psychological experience of perceived crowding (Stokols, 1972). Indeed, crowding can be measured objectively as people per unit of space, termed social density, or as people per metre per minute, termed pedestrian flow. In parallel, subjective assessments such as surveys can help gauge how people perceive and experience various levels of social density. Furthermore, perceptions of density and crowding appear malleable to a certain extent. The psychological perception of when a space feels crowded is not a stable construct based on the number of other people but is influenced by other personal, spatial, and behavioural parameters, from noise to room height. For instance, does a given number of

people per space feel the same in the double-height space of a library reading room or on the platform of a busy train station? Perceptions of crowding can be moderated by internal factors such as personality traits (Wiesenfeld, 1987), demographics (Yin et al., 2020), and social culture (e.g., contact vs noncontact cultures; Kaya & Weber, 2003). Environmental factors can also influence how social density is perceived, from neighbourhood density (Bonnes et al., 1991) to design (Evans et al., 1996). Indeed, our recent findings suggest that similar numbers in pedestrian crowds lead to different evaluations depending on whether these are experienced indoors or outdoors (Mavros et al., in review). Perhaps acknowledging the importance of social presence for urban affect, even architectural visualization is moving away from older paradigms of showcasing architecture without people; new rendering software includes increasingly realistic and often animated humans engaging in various activities from walking to jogging and street-dancing as a means to help stakeholders better imagine the experience of spaces as they would be built.

THE NEUROSCIENCE OF PEOPLE-ENVIRONMENT INTERACTIONS

Following the age of sanitary reform, post-industrial cities underwent a huge transformation towards provision of healthier living spaces. The focus was then placed on physical health and decreasing the spread of communicable diseases. Better ventilation in buildings and improvements in sewer drainage, water supply, and garbage removal led to major changes in the planning of urban neighbourhoods (Peterson, 1979). Another challenge to well-being resulting from city life recognized back then was cognitive overstimulation. As Frederick Law Olmsted, the 19th-century designer of New York's Central Park, wrote: 'A man's eyes cannot be as much occupied as they are in large cities by artificial things [...] without harmful

effect, first on his mental and nervous system and ultimately on his entire constitutional organisation' (quoted in Beveridge, 1995, p. 36). Planners in this era sought to alleviate the overwhelming crowdedness by introducing nature into cities through 'breathing spaces': parks and outdoor recreation areas adjacent to working-class neighbourhoods (Corburn, 2007). The large open areas there promoted ventilation, and vegetation captured dust and toxins that would otherwise be inhaled by residents. In spite of clear improvements in living standards and lowering the burden of many dangerous infectious diseases through improved sanitation, the efforts of 19th century urban planners to alleviate the negative impact of post-industrial cities on mental health were unsuccessful.

Indeed, despite the clear improvement in living standards in cities across the world since the 19th century, although this admittedly varies across neighbourhoods and cities, the quality of our urban living environment is still, in many ways, innately defective. Studies consistently show that mental disorders such as depression, anxiety, and substance dependence are as much as 38 per cent more prevalent in urban populations than rural ones (for example see a meta-analysis of 20 studies by Peen et al., (2010)). The COVID-19 pandemic spurred increasing discussion on mental health in cities and how contact with urban nature helped keep people sane through all the associated hardships. Widely introduced lockdowns led to sensory deprivation from a limited diversity of environmental exposures, including exposure to nature; this drew attention by analogy to prison research. These studies have documented that living in confinement deprived of diverse stimuli for prolonged periods of time can lead to psychological distress (Hocking, 1970; Leiderman, 1962; Haney, 2003). As a potential alleviating factor, green space accessibility has been shown to be very important for city residents as a way to alleviate the stress and uncertainty arising during the pandemic (Da

“Aesthetic and other elements of the architectural environment interact with levels of social density to influence the subjective experience of urban spaces.”

Schio et al., 2021; Vos et al., 2022). Such findings are a call for architects and urban planners to enquire how the planning and design of our cities and buildings contribute to social and psychological issues and what can be done to heal contemporary cities.

Thanks to years of research in environmental psychology, today, a consensus has emerged among scientists that urban green spaces are media of sensitive and restorative values; the consensus is corroborated by a growing body of empirical and observational knowledge (van den Berg et al., 2007; Berman et al., 2008; Hartig et al., 2003; Olszewska-Guizzo et al., 2022). Several theories have been proposed to describe the mechanisms by which green spaces in cities can benefit people’s well-being; such theories are typically centred around visual properties. The biophilia hypothesis postulates that all humans are innately affiliated with any natural elements due to natural asymmetry (fractal patterns) in nature (Kellert & Wilson 1993). The attention restoration theory (ART) argues that urban space provides sensory overstimulation and constant ‘goal-oriented’ patterns of attention, which cause mental fatigue, whereas natural environments recruit the opposite pattern of ‘stimuli-driven’ attention, which allows restoration of our depleted capacity for attention (Kaplan, 2001). According to ART, contact with nature induces ‘soft fascination’, which is key to the restorative experience (Kaplan & Berman, 2010). Restorative experience is the state in which the environment gently engages the individual’s attention, diffusing and relaxing strenuous cognitive processing and focus on detail. However, it is still not well understood which are the specific components of the environment that can foster positive health outcomes, either from passive or through active interaction with them.

Here, environmental neuroscience with portable, wireless, yet reliable brain imaging tools can help unveil some of the unconscious mechanisms by which the human brain operates in various

spaces. During the last decade, research in environmental neuroscience outside the laboratory has been spurred by new mobile electroencephalography (EEG) technologies such as wireless and ambulatory signal acquisition devices. New developments in signal processing techniques have allowed the effects of walking and other muscle artifacts to be minimized and removed from the EEG signal, which has enabled the study of ‘cognition in action’ (Gramann et al., 2014, p. 22), in many fields from sports science to pedestrian navigation, environmental effect, and psychological restoration, recordings have been validated outside the laboratory for studying both cognitive tasks and person–environment interactions (Aspinall et al., 2015; Mavros, 2019). It is possible to compare urban with rural and to investigate responses to specific features within built-up and green spaces. We can learn how certain design strategies can impact the emotions, cognitive and attention processes, and even social interactions, including navigating through crowds: which spaces make us more nervous and which calm us, and finally, how this all applies to the specific groups in the society, for example the elderly, youth, and others.

THE NEUROSCIENCE OF AESTHETIC EXPERIENCE

The aesthetic experience of art or space engages our cognitive and emotional processes. This is the focus of neuroaesthetics, a young and fast-growing field concerned with the biological and neural basis of aesthetic experience and judgement. For instance, behavioural data show that artworks that are perceived as being more ugly or more beautiful at either extreme of an aesthetic judgement scale are remembered better. Similarly, the aesthetic quality of urban spaces can influence route choice (Johansson et al., 2016). However, Bourdieu’s (1977) idea that the perception of art is socially constructed is still debated: whether beauty is inherent to the object of appreciation or to the person contemplating it. Several studies suggest that certain

physical features of an artwork, such as symmetry and contrast, influence aesthetic rating. Other arguments propose that aesthetic preferences are related to the visual complexity of objects and to scenes that progressively reveal visual variations at multiple scales, which has been called the fractal dimension (Taylor, 2021). Here, an interesting parallel can be observed. Loss of complexity in the fractal dimension of physiological signals is associated with a wide range of physical and psychological disorders. Similarly, people are more attracted to space and objects with higher complexity. Other recent studies have also demonstrated a connection between one's sense of self and past experience and aesthetic judgement (Lee et al., 2020).

The links between neuroscience and architecture are also strengthening, as in the emerging fields of neuroarchitecture and neurourbanism, as part of renewed interest in the links between environment and mental health. This is partly due to our scientific curiosity to understand how our brains perceive and react to different environments, but equally to advancing evidence-based urban policy, planning, and design. Psychophysiological measurement methods such as eye-tracking, EEG, and fMRI have been used to study the affective and attentional processes of looking at urban or natural environments. Eye-tracking metrics such as dwell time can highlight, for instance, how the facades of buildings attract visual attention (Hollander et al., 2020), and EEG measures can reveal people's emotional reactions, such as approach/positive emotion (e.g., frontal-alpha asymmetry), cognitive load (e.g., theta/beta ratio), and more.

The framework of neuroaesthetics has been applied to architecture and urban and landscape design. Researchers have started to systematically examine people's responses to architectural spaces, both conscious responses through self-reports and unconscious ones through such psychophysiological measures as skin conductance and brain

activity. These studies can provide insights into the influence of factors such as form, proportions, height, materiality, and other factors on experience, attention, and emotion (Vartanian et al., 2015).

Coburn et al. (2020) identified three components driving our perception and appreciation of interior scenes: coherence, fascination, and hominess (see Chatterjee et al., 2021). This last is a dimension specific to architectural interiors that represents the extent to which a space feels comfortable and personal. The first two dimensions have also been found to be critical axes in our perception of natural scenes. Coherence captures the degree to which a space feels organized to the viewer, and fascination encapsulates the visual richness and complexity of a space. Interestingly, the research found that fascination was associated with a viewer's sense of excitement and desire to explore the space. At the urban scale, fascination might be related with the notion of urban vibrancy (Montgomery, 1998).

Within the fields of restorative theory and landscape architecture, the contemplative landscape model (CLM; Olszewska-Guizzo, 2018) is a similar attempt to establish a universal framework for design and assessment of urban green spaces. It consists of capturing seven key-characteristics of a landscape scene: layers, landform, biodiversity, colour and light, compatibility, archetypal elements and character of peace, and silence. From a phenomenological standpoint, landscape views are just images, perceived by senses and processed by the brain. Recent studies on highly contemplative spaces in Singapore have shown that spaces ranked highly using the CLM framework can elicit changes in patterns of brain activity associated with relaxation and mindfulness (Olszewska-Guizzo et al., 2020) and induce the brain activity desirable in clinical depression treatment (Olszewska-Guizzo et al., 2022).

Urban design and architecture have a long tradition of systematically measuring objective factors relating to form and

Figure 2 →
Example of neuropsychological measurement in a field experiment: a participant wearing a portable multimodal fNIRS/EEG headset while exposed to a dense urban environment (photograph by Agnieszka Olszewska-Guizzo).



function. Capturing subjective factors such as aesthetic and design qualities present various challenges, and for some they are indeed ‘unmeasurable’ (Ewing & Clemente, 2013), but they are nonetheless important to understanding how people experience and engage with the urban and architectural space. Factors such as enclosure, imageability, and legibility cannot be as readily assessed building or room height but are linked with human behaviour such as walkability and urban vitality.

In most of the studies mentioned above, the presence of others is considered a confound, something to be experimentally controlled for: stimuli are created carefully to not include people because these could bias the results. While the research on how these parameters influence spatial experiences is ongoing, it is equally important to understand how it is also modulated by human presence; whether scenic or dull, streets do not feel the same when they are empty as when bustling with activity.

THE NICE PROJECT

We recently began research on the NICE project to investigate these questions. What is the joint influence of environmental qualities and social copresence? Are the levels when a space starts to feel crowded stable, or do they depend on aesthetic and other spatial qualities?

Our approach is to combine the methodologies discussed above into environmental neuroaesthetics. In our studies, we show participants urban environments that vary in design, aesthetics, and spatial properties as well as levels of human occupancy. During these experiments, we record their physiological arousal, brain activity, and conscious preferences. Over the next year, we will employ this method to conduct a series of online, laboratory, and field studies to understand the joint influence of social density and environmental aesthetics on perceptions of crowding, sense of presence, and willingness to visit.

Our research also raises new questions. How will environmental neuroscience relate to our notions of designers’ agency, intuition, and preference? The psychological effects of space on well-being are increasingly recognized. This approach will allow us to better understand design paradigms. Arguably, our methodology can only capture instantaneous responses to the environment, but this can complement other longitudinal research on the effects of prolonged exposure to various environmental factors.

Our efforts aim to redefine our perception of human presence in spaces. Rather than simply considering what is the maximum capacity when designing spaces, perhaps it is time to consider optimal capacity too – the minimum presence to make a space feel vibrant and attractive. Furthermore, if there is a link between the aesthetics of the environments and the acceptance or even attraction of higher number of people (i.e. tolerance for higher crowding), this can also have implications for design approaches of spaces, who due to their function or location in the city, are expected to be bustling with crowds.

CONCLUSION

Inspired by the experience of emptying cities during the pandemic, we argue that it is important to re-examine notions of social density and crowding with novel methods from neuroscience and environmental psychology. Aesthetic and other elements of the architectural environment interact with levels of social density to influence the subjective experience of urban spaces. Establishing an empirical understanding of the importance of both well-designed and socially vibrant environments will not only increase our knowledge of how people engage with their existing environments, but help drive better policy, planning, and design decisions.



Figure 3 ▶ p.151

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Figure 1
Image of Orchard Road, Singapore's prime retail and pedestrian avenue, during pre-pandemic pedestrian rush-hour. How does the number of people walking influence our subjective experience of the environment? (photograph by Panagiotis Mavros.)

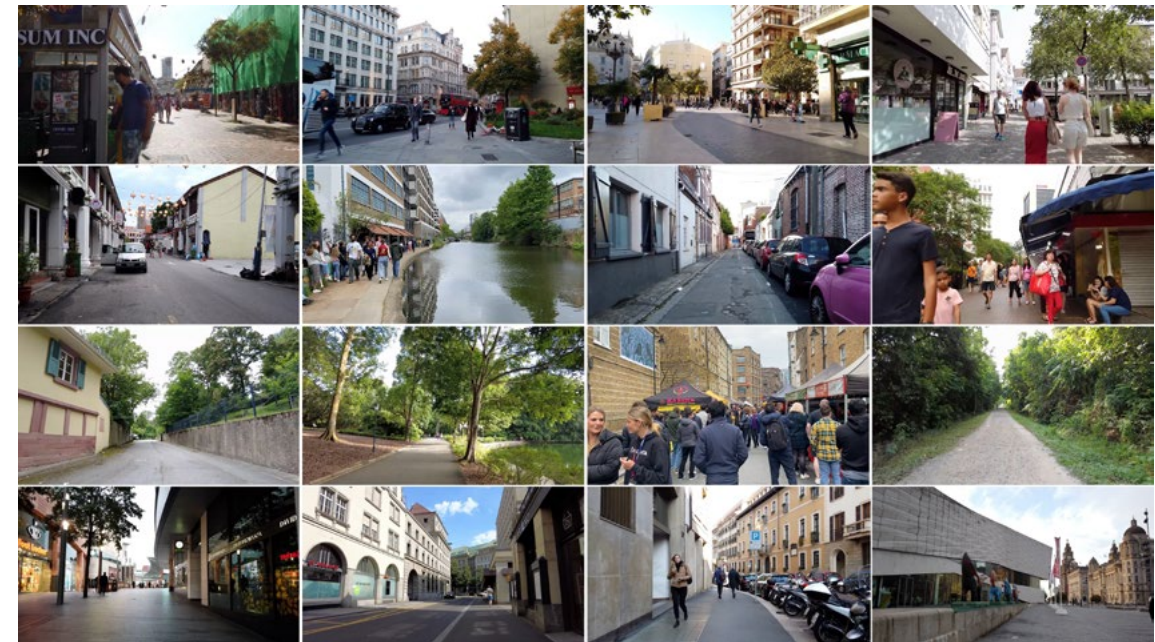


Figure 3
Frame samples from video walkthroughs that were collected from different cities and countries. These videos will serve as a large database of environmental stimuli, varying in environmental typology, aesthetic qualities and social pedestrian density. They will be used to assess subjective experience and environmental preferences (photographs by the authors, adapted from original video sources by Daniel Szczepansky (POP Travel) CC BY SA and Thomas Lim).